

CAPSDAC Level 3 MLE Forecast Platform

Synthetic public demo: 3-month-ahead enrollment, capacity risk, staffing need, model registry, and monitoring

Selected Model

linear_regres...

RMSE0.378

MAE0.308

MAPE0.348%

Model Leaderboard

	model_name	params	mae	rmse	mape	r2
13	extra_trees_lightgbm_style	{'max_depth': 6, 'min_samples_leaf': 2, 'n_estimators': 80}	4.6428	5.2638	5.1224	0.8659
14	extra_trees_lightgbm_style	{'max_depth': 6, 'min_samples_leaf': 1, 'n_estimators': 80}	4.5208	5.2819	4.9637	0.8657
15	random_forest	{'max_depth': 10, 'min_samples_leaf': 1, 'n_estimators': 80}	4.5888	5.2867	5.0759	0.8632
16	gradient_boosting	{'learning_rate': 0.03, 'max_depth': 3, 'n_estimators': 80}	4.5058	5.317	4.7398	0.8612
17	random_forest	{'max_depth': 6, 'min_samples_leaf': 1, 'n_estimators': 80}	4.6156	5.3325	5.0939	0.8603
18	random_forest	{'max_depth': 6, 'min_samples_leaf': 1, 'n_estimators': 160}	4.5595	5.3408	5.033	0.86
19	random_forest	{'max_depth': 6, 'min_samples_leaf': 2, 'n_estimators': 160}	4.7044	5.4327	5.1531	0.8538
20	random_forest	{'max_depth': 6, 'min_samples_leaf': 2, 'n_estimators': 80}	4.7799	5.4626	5.246	0.8525
21	gradient_boosting	{'learning_rate': 0.03, 'max_depth': 2, 'n_estimators': 80}	5.5043	6.3121	5.6765	0.8019
22	median_baseline	{}	14.7778	18.2398	15.141	-0.5834

Site Forecasts

	ForecastMonth	ForecastHorizonMonths	VendorNumber	VendorName	PreschoolCDSCode	SiteName	County	FundingTyp
0	2025-12	3	2841	Valley Early Learning	19648896000004	Valley Early Learning LA	Los Angeles	CSPP
1	2025-12	3	2841	Valley Early Learning	19648896000005	Valley Early Learning Pomona	Los Angeles	CCTR
2	2025-12	3	4312	Golden State Preschool	37683386000003	Golden State San Diego	San Diego	CCTR
3	2025-12	3	5920	Central Coast Kids	40688016000006	Central Coast Kids SLO	San Luis Obispo	CSPP
4	2025-12	3	6785	BrightStart Education	19647336000001	BrightStart Irvine	Orange	CSPP
5	2025-12	3	6785	BrightStart Education	19647336000002	BrightStart Anaheim	Orange	CSPP

Drift Monitoring

```

{
  "baseline_end" : "2024-12-01"
  "current_start" : "2025-07-01"
  "method" : "population_stability_index"
  "thresholds" : {
    "stable" : "<0.10"
    "moderate" : "0.10-0.25"
    "high" : ">=0.25"
  }
  "features" : [
    0 : {
      "feature" : "lag_1"
      "psi" : 2.3597096711324714
      "status" : "high"
      "baseline_mean" : 80.81944444444444
      "current_mean" : 83.33333333333333
    }
    1 : {
      "feature" : "lag_3"
      "psi" : 3.2726162073778817
    }
  ]
}

```

Champion Registry

```

{
  "registry_version" : "v3-local-registry-1"
  "updated_utc" : "2026-06-16T17:39:56.219521+00:00"
  "decision" : "promoted"
  "decision_reason" :
    "Challenger RMSE is within acceptance threshold versus
    existing champion."
  "champion_model_uri" :
    "models/registry/champion_model.joblib"
  "challenger_model_uri" :
    "models/registry/latest_challenger_model.joblib"
  "champion_run_id" : "capdac-v3-20260616173956-80705b"
  "champion_model_name" : "linear_regression"
  "champion_metrics" : {
    "mae" : 0.30839270729783247
    "rmse" : 0.37777862424033093
    "mape" : 0.3481107428700832
    "r2" : 0.9991995200142202
  }
  "latest_challenger" : {

```

```
"status" : "high"
"baseline_mean" : 79.93055555555556
"current_mean" : 78.83333333333333
}
▼ 2 : {
  "feature" : "lag_6"
  "psi" : 3.7575508923896996
  "status" : "high"
  "baseline_mean" : 78.56944444444444
  "current_mean" : 84.66666666666667
}
▼ 3 : {
  "feature" : "lag_12"
  "psi" : 4.719609933110805
  "status" : "high"
  "baseline_mean" : 75.83333333333333
  "current_mean" : 82.11111111111111
}
▼ 4 : {
  "feature" : "rolling_3"
  "psi" : 1.1216762500206303
  "status" : "high"
  "baseline_mean" : 80.37962962962963
  "current_mean" : 80.83333333333334
}
▼ 5 : {
  "feature" : "rolling_6"
  "psi" : 5.04495120298198
  "status" : "high"
  "baseline_mean" : 79.70370370370371
  "current_mean" : 81.33333333333331
```

```
"run_id" : "capsdac-v3-20260616173956-80705b"
"model_name" : "linear_regression"
▼ "metrics" : {
  "mae" : 0.30839270729783247
  "rmse" : 0.37777862424033093
  "mape" : 0.3481107428700832
  "r2" : 0.9991995200142202
}
}
```

```
}
  6 : {
    "feature" : "rolling_12"
    "psi" : 6.336285394850933
    "status" : "high"
    "baseline_mean" : 78.33564814814814
    "current_mean" : 84.47222222222223
  }
  7 : {
    "feature" : "month_sin"
    "psi" : 8.593387339706291
    "status" : "high"
    "baseline_mean" : -8.63979223605633e-18
    "current_mean" : -0.7886751345948126
  }
  8 : {
    "feature" : "month_cos"
    "psi" : 8.593387339706291
    "status" : "high"
    "baseline_mean" : -4.5726687344757515e-17
    "current_mean" : -0.45534180126147983
  }
  9 : {
    "feature" : "trend_index"
    "psi" : 11.561538276029477
    "status" : "high"
    "baseline_mean" : 18.5
    "current_mean" : 32
  }
  10 : {
    "feature" : "site_capacity"
    "psi" : 0
```

```
"status" : "stable"
"baseline_mean" : 102.1666666666667
"current_mean" : 102.1666666666667
}
▼ 11 : {
  "feature" : "capacity_utilization_lag1"
  "psi" : 2.955781145280863
  "status" : "high"
  "baseline_mean" : 0.7861364689837149
  "current_mean" : 0.8113312556477682
}
▼ 12 : {
  "feature" : "attendance_rate_lag1"
  "psi" : 4.261492107662543
  "status" : "high"
  "baseline_mean" : 0.9133333333333334
  "current_mean" : 0.899673079295489
}
▼ 13 : {
  "feature" : "teacher_student_ratio_lag1"
  "psi" : 2.001370727318176
  "status" : "high"
  "baseline_mean" : 0.08979604355297605
  "current_mean" : 0.088275264603175
}
▼ 14 : {
  "feature" : "county_population_index"
  "psi" : 0.09241962407465937
  "status" : "stable"
  "baseline_mean" : 108
  "current_mean" : 109.5
```

```

    }
    15 : {
      "feature" : "county_unemployment_proxy"
      "psi" : 6.589473454185372
      "status" : "high"
      "baseline_mean" : 4.375
      "current_mean" : 4.300000000000001
    }
    16 : {
      "feature" : "EnrollmentCount"
      "psi" : 3.697291904538945
      "status" : "high"
      "baseline_mean" : 81.27777777777777
      "current_mean" : 87.33333333333333
    }
  ]
  "overall_status" : "high"

```

Retraining Decision

```

{
  "retraining_policy" : "monthly_challenger_training_with_champion_comparison"
  "triggered" : true
  "trigger_reasons" : [
    0 : "Data drift status is high."
  ]
  "new_month_workflow" : [
    0 : "ingest_new_snapshot"
    1 : "validate_data_contracts"
    2 : "rebuild_feature_store"
    3 : "train_challenger_models"
    4 : "compare_against_champion"
    5 : "promote_or_reject"
  ]
}

```

```
    6 : "publish_forecasts_and_monitoring_report"  
  ]  
  "production_note" :  
  "In production this would be scheduled by Airflow/Cloud Composer or GitHub Actions and registered in MLflow/Vertex  
  AI."  
}
```